

COURSE TITLE : COURSE CODE : CSE211

PMT No.: 2272CSE101005
Paper Code : A

Time Allowed: 01:30 hrs.

COMPUTER ORGANIZATION AND DESIGN

Max. marks: 10

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- 1) The most significant application of flip - flop are :
(a) counters (b) shift registers (c) transfer register (d) all of these
- 2) Flip - flop outputs are always :
(a) complimentary (b) the same (c) independent of each other (d) same as inputs
- 3) In which of the following gates, the output is 1, if and only if at least one input is 1?
(a) NOR (b) AND (c) OR (d) NAND
- 4) An OR gate can be imagined as
(a) switches connected in series (b) switches connected in parallel (c) MOS transistors connected in series (d) none of these
- 5) What is the minimum number of two-input NAND gates used to perform the function of two input OR gate?
(a) one (b) two (c) three (d) four
- 6) Which of the following gates are added to the inputs of the OR gate to convert it to the NAND gate?
(a) NOT (b) AND (c) OR (d) XOR
- 7) Digital circuits easier to design than analog circuits because
(a) they do not control electricity precisely over a wide range. (b) they are made in the form of ICs. (c) all elements of digital circuits are from the same family. (d) they are smaller in size.
- 8) Consider the following statements:
1. Taking 2's complement is equivalent to sign change.
2. In the 2's complement representation the most significant bit (MSB) is zero for a positive number.
3. In a 4 bit binary representation of a binary number A, $A + 1$'s complement of $A = 24$.
Which of the above statements are correct?
(a) 1 and 2 only (b) 1 and 3 only (c) 2 and 3 only (d) 1, 2 and 3
- 9) Which of the following counters can be used to divide the clock frequency of a microprocessor by 5?
(a) 3 bit counter (b) 5 bit counter (c) mod 3 counter (d) mod 5 counter
- 10) Register renaming is done in pipelined processors
(a) as an alternative to register allocation at compile time (b) for efficient access to function parameters and local variables (c) to handle certain kinds of hazards (d) as part of address translation
- 11) The CPU of a computer takes instruction from the memory and executes them. The phases of instruction cycle are
(a) load cycle (b) time sequence (c) fetch-execute cycle (d) clock cycle
- 12) Both the arithmetic logic unit (ALU) and control section of CPU employ special purpose storage locations called
(a) decoders (b) buffers (c) multiplexer (d) registers
- 13) The register which keeps track of the execution of a program and which contains the memory address of the instruction currently being executed is called
(a) index register (b) memory address register (c) program counter (d) instruction register

14) The register used as a working area in CPU is
 (a) program counter (b) instruction register (c) instruction decoder (d) accumulator

15) The hardware in which data may be stored for a computer system is called
 (a) register (b) memory (c) chip (d) peripheral

16) The program below uses six temporary variables a, b, c, d, e, f.

a = 1

b = 10

c = 20

d = a + b

e = c + d

f = c + e

b = c + e

e = b + f

d = 5 + e

return d + f

Assuming that all operations take their operands from registers, what is the minimum number of registers needed to execute this program without spilling?

(a) 2 (b) 3 (c) 5 (d) 6

17) The content of which of the following determines the state of the CPU at the end of the execute cycle (when the interrupt is recognized)?

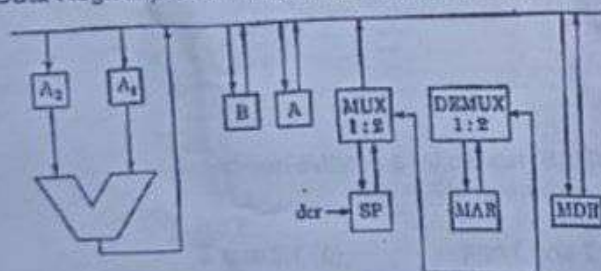
1. Program counter
2. Processor register
3. Certain status conditions

Select the correct answer using the codes given below:

Codes:
 (a) 1 and 2 (b) 2 and 3 (c) 1 and 3 (d) 1, 2 and 3

18) The unit that is used to supervise each instructions in the CPU is
 (a) Control register (b) Control logic unit (c) ALU (d) Address register

19) Consider the following data path of a simple non-pipelined CPU. The registers A, B, A₁, A₂, MDR, the bus and the ALU are 8-bit wide. SP and MAR are 16-bit registers. The MUX is of size 8 × (2 : 1) and the DEMUX is of size 8 × (1 : 2). Each memory operation takes 2 CPU clockcycles and uses MAR (Memory Address Register) and MDR (Memory Data Register). SP can be decremented locally.



The CPU instruction "push r", where r = A or B, has the specification

$M[SP] \leftarrow r$

$SP \leftarrow SP - 1$

How many CPU clock cycles are needed to execute the "push r" instruction?

(a) 2 (b) 3 (c) 4 (d) 5

20) Which of the following is/are true of the auto increment addressing mode?

1. It is useful in creating self relocating code
2. If it is included in an Instruction Set Architecture, then an additional ALU is required for effective address calculation
3. The amount of increment depends on the size of the data item accessed.

(a) 1 only (b) 2 only (c) 3 only (d) 2 and 3 only

21) The register which holds the address of the location to or from which data are to be transferred is called
 (a) index register (b) instruction register
 (c) memory address register (d) memory data register

22) The total size of address space in a virtual memory system is limited by
 (a) the length of MAR (b) the available secondary storage
 (c) the available main memory (d) All of the above

23) A 4-stage pipeline has the stage delays as 150, 120, 160 and 140 nanoseconds respectively. Registers that are used between the stages have a delay of 5 nanoseconds each. Assuming constant clocking rate, the total time taken to process 1000 data items on this pipeline will be

- (a) 120.4microseconds (b) 160.5microseconds (c) 165.5microseconds (d) 590.0 microseconds

24) The Register which keeps track of the execution of a program and which contains the memory address of the next instruction to be executed is called

- (A) Instruction register (B) Program counter (C) Index register (D) Memory address register

25) When a subroutine is called the address of the instruction following the *CALL* instruction stored in the

- (A) Stack (B) Accumulator (C) Program counter (D) Stack pointer

26) A CPU has 24-bit instructions. A program starts at address 300 (in decimal). Which one of the following is a legal program counter (all values in decimal)?

- (a) 400 (b) 500 (c) 600 (d) 700

27) A system of letters, numbers symbols adopted by computer manufacturer as an abbreviated form of instruction sets is called

- (a) mesh (b) monitor (c) modern (d) mnemonic

28) A micro-programmed control unit

- (a) is faster than a hard-wired control unit (b) facilitates easy implementation of new instructions
(c) is usefull when very small programs are to be run (d) usually refers to the control unit of a microprocessor

29) Match the following:

List I	List II
P. Indirect Addressing	1. Loops
Q. Auto decrement Addressing	2. Constants
R. Immediate Addressing	3. Pointers

- (a) P - 1, Q - 3, R - 2 (b) P - 3, Q - 1, R - 2 (c) P - 2, Q - 1, R - 3 (d) P - 3, Q - 2, R - 1

30) The use of multiple register windows with overlap causes a reduction in the number of memory accesses for

1. function locals and parameters
2. register saves and restores
3. instruction fetches

- (a) 1 only (b) 2 only (c) 3 only (d) 1, 2 and 3

- End of Question Paper -

- 9) Which of the following statements about relative addressing mode is FALSE?
- It enables reduced instruction size
 - It allows indexing of array elements with same instruction
 - It enables easy relocation of data
 - It enables faster address calculations than absolute addressing
- 10) An instruction used to set the carry flag in a computer can be
- Data control
 - Process control
 - Logical
 - Data transfer
- 11) The unit that is used to supervise each instructions in the CPU is
- Control register
 - Control logic unit
 - ALU
 - Address register
- 12) The performance of a pipelined processor suffers if
- the pipeline stages have different delays
 - consecutive instructions are dependent
 - the pipeline stages share hardware resources
 - All of the above
- 13) In which of the following gates, the output is 1, if and only if at least one input is 1?
- NOR
 - AND
 - OR
 - NAND
- 14) When an inverter is placed between both inputs of an SR flip-flop, then resulting flip-flops
- JK flip-flop
 - D flip-flop
 - T flip-flop
 - Master slave
- 15) An OR gate can be imagined as
- switches connected in series
 - switches connected in parallel
 - MOS transistors connected in series
 - none of these
- 16) A processor that has carry, overflow and sign flag bits as part of its program status word (PSW) part of the following two 2's complement numbers 01001101 and 11101001. After the execution of this addition the status of the carry, overflow and sign flags, respectively will be:
- 1, 1, 0
 - 1, 0, 0
 - 0, 1, 0
 - 1, 0, 1
- 17) A basic instruction that can be interpreted by a computer generally has
- an operand and an address
 - a decoder and an accumulator
 - sequence register and decoder
 - an address and decoder
- 18) The function of program counter (PC) holds
- temporary
 - address for memory
 - memory operand
 - address for
- 19) A 4-stage pipeline has the stage delays as 150, 120, 160 and 140 nanoseconds respectively. Register used between the stages has a delay of 5 nanoseconds each. Assuming constant clocking rate, the time to process 1000 data items on this pipeline will be
- 120.4 microseconds
 - 160.5 microseconds
 - 165.5 microseconds
 - 590.0 microseconds
- 20) In case the code is position dependent, the most suitable addressing mode is
- Direct mode
 - Indirect mode
 - Relative mode
 - Indexed mode
- 21) Which one of the following addressing modes is used in the instruction PUSH B?
- Direct
 - Register
 - Register indirect
 - Immediate
- 22) Which of the following gates would give output 1 when one input is 1 and other input is 0?
- OR gate
 - AND gate
 - NAND gate
 - both (a) and (b)
- 23) Which of the following gates is known as coincidence detector?
- AND gate
 - OR gate
 - NOT gate
 - NAND gate
- 24) Popular application of flip-flop are
- counters
 - shift registers
 - transfer register
 - all of these
- 25) Which of the following interrupts are unmaskable interrupts?
- RST 5.5
 - RST 7.5
 - TRAP
 - INTR
- 26) Hardware unit which is used to monitor computer processing is
- console
 - dot matrix printer
 - mouse
 - ROM

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- 1) Which of the following counters can be used to divide the clock frequency of a microprocessor by 5?
- (a) 3 bit counter (b) 5 bit counter (c) mod 3 counter (d) mod 5 counter

- 2) The control unit of computer
- (a) performs ALU operations on the data (b) controls the operation of the output devices
- (c) is a device for manually operating the computer (d) directs the other unit of computers

- 3) Consider the following statements regarding registers and latches:

1. Registers are made of edge-triggered FFs, whereas latches are made from level-triggered FFs.
2. Registers are temporary storage devices whereas latches are not.
3. A latch employs cross-coupled feedback connections.
4. A register stores a binary word whereas a latch does not.

Which of the statements given above are correct?

- (a) 1 and 2 (b) 1 and 3 (c) 2 and 3 (d) 3 and 4

- 4) Consider the following program segment. Here R1, R2 and R3 are the general purpose registers. Instruction Operation Instruction size (no. of words)

MOV R1, (3000)	R1 ← M[3000]	2
LOOP		
MOV R2, M[R3]	R2 ← M[R3]	1
ADD R2, R1	R2 ← R1 + R2	1
MOV R3, R2	M[R3] ← R2	1
INC R3	R3 ← R3 + 1	1
DEC R1	R1 ← R1 - 1	1
BNZ LOOP	Branch on not zero	2
HALT		Stop

Assume that the content of memory location 3000 is 10 and the content of the register R3 is 2000. The content of the memory locations from 2000 to 2010 is 100. The program is loaded from the memory location 10. numbers are in decimal.

Assume that the memory is byte addressable and the word size is 32 bits. If an interrupt occurs during the instruction "INC R3", what return address will be pushed on to the stack?

- (a) 1005 (b) 1020 (c) 1024 (d) 1040

- 5) In a microprocessor, the register which holds the address of the next instruction to be fetched is:
- (a) accumulator (b) program counter (c) stack pointer (d) instruction register

- 6) Which of the following is a set of general purpose internal registers?
- (a) stack (b) scratch pad (c) address register (d) status register

- 7) In indexed addressing mode with scaling, the effective address is calculated as
- (a) Index + scaling + signed displacement (b) (Index * scaling) + signed displacement
- (c) Index + (scaling * displacement) (d) (Index + scaling) * signed displacement

- 8) The address of the location to or from which data are to be transferred is called

- (A) Memory data register
- (B) Memory address register
- (C) Program counter
- (D) Index register

27) A microprocessor has 24 address lines and 32 data lines. If it uses 10 bits of op code, the size of its Memory Buffer register is

- (a) 22 bits (b) 24 bits (c) 32 bits (d) 14 bits

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29) For computers based on three-address instruction formats, each address field can be used to specify which of the following

- S1: A memory operand
S2: A processor register
S3: An implied accumulator register
(a) Either S1 or S2 (b) Either S2 or S3 (c) Only S2 and S3 (d) All of S1, S2 and S3

30) The main difference (a) between a CISC and a RISC processor is that a RISC processor typically

- (a) has fewer instructions and addressing modes (b) has more registers
(c) is easier to implement using hard-wired control logic (d) all of these

= End of Question Paper =

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- 6) The CPU of a computer takes instruction from the memory and executes them. This process is called
 - (a) load cycle
 - (b) time sequence
 - (c) fetch-execute cycle
 - (d) clock cycle
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```

a = 1
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c = 20
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e = c + d
f = c + e
b = c + e
e = b + f
d = 5 + e
return d + f

```

Assuming that all operations take their operands from registers, what is the minimum number of registers needed to execute this program without spilling?

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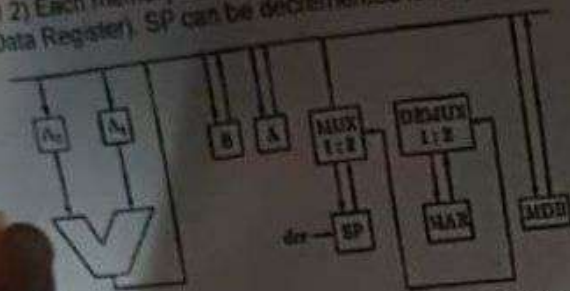
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The CPU instruction "push r", where r = A or B, has the specification

M[SP] ← r
SP ← SP - 1

How many CPU clock cycles are needed to execute the "push r" instruction?

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